

Peoples Bakers — Project Guide

Frontend + Backend architecture, server run කරන හැටි, bug fixes, mobile responsive, සහ daily use — සම්පූර්ණ සිංහල මාර්ගෝපදේශය

Version: 1.1 | **Date:** June 2026 | **Stack:** HTML/CSS/JS + Node.js 22+ + SQLite

මේ guide එක කවුද? — Project එක internet hosting කරන්න කලින්, local computer එකේ run කරලා understand කරගන්න ඕන අයට. Developer නොවුනත් පියවර follow කරලා site එක test කරන්න පුළුවන්.

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1. Project එක මොකක්ද?

Peoples Bakers website එක Sri Lankan bakery business එකක් online වෙන්න build කරපු full-stack web application එකක්. Customer එකට:

- Products & cakes browse කරන්න
- Shopping basket එකට items add කර order place කරන්න
- WhatsApp / phone / PayHere payment use කරන්න
- Account create කර order history බලන්න
- Feedback send කරන්න
- Outlets map එකෙන් branch සොයන්න

Business owner / staff ට **Admin panel** එකෙන් orders, products, feedback manage කරන්න පුළුවන්.

Important: මේ project එක *plain HTML file double-click* කරලා open කරන්න බෑ. **Node.js server** එක run කරලා `http://localhost:3000` වලින් open කරන්න ඕන. Live Server, file://, හෝ PHP-only hosting වල backend work වෙන්වේ නැහැ.

2. Frontend vs Backend — simple විස්තරය

Frontend (ඉදිරිපස — user දකින එක)

Browser එකේ run වෙන දේවල්. User click කර, scroll කර, form fill කරන සියල්ල.

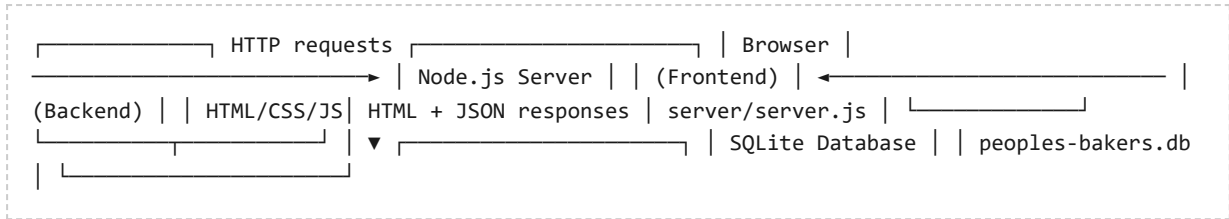
- **HTML** — page structure (index.html, order.html, ...)
- **CSS** — colors, layout, mobile design (css/style.css)
- **JavaScript** — cart, checkout, language switch, maps (js/ folder)
- **Images** — products, hero photos (images/ folder)

Frontend එක alone වලින් orders save වෙන්නේ, login work වෙන්නේ, admin panel live data show වෙන්නේ නැහැ. API call කරන්න backend server එක ඕන.

Backend (පසුපස — server එකේ run වෙන එක)

Computer/server එකේ run වෙන **Node.js + Express** program එක (server/server.js). මේක:

- Static files (HTML, CSS, JS, images) browser එකට serve කරනවා
- REST API endpoints handle කරනවා (/api/...)
- Login tokens (JWT) verify කරනවා
- Orders, users, feedback, products **SQLite database** එකේ save කරනවා
- PayHere payment webhooks process කරනවා
- Admin-only data protect කරනවා



Simple analogy: Frontend = shop display window. Backend = kitchen + cashier + record book.
 Customer window එක බලනවා; order kitchen එකට යනවා database එකේ save වෙනවා.

3. Technology stack

Layer	Technology	Purpose
Frontend	HTML5, CSS3, Vanilla JavaScript	Pages, styling, cart, i18n
Backend	Node.js 22.5+ , Express 4	API + static file server
Database	SQLite (node:sqlite built-in)	Users, orders, products, feedback
Auth	JWT + httpOnly cookies, bcrypt	Secure login
Payments	PayHere integration	Online card/LankaQR payments
Email	Nodemailer (optional SMTP)	Order confirmations, welcome emails
PWA	service-worker.js	Offline cache for static assets
i18n	js/i18n.js + en/si/ta JSON	3 languages

4. Folder structure — files කොහෙද?

```
Peoples-Bakers-master/
├─ index.html          ← Home page
├─ order.html          ← Order online + checkout
├─ products.html       ← Products & cakes catalogue
├─ account.html        ← Customer account
├─ login.html          ← Sign in / register
├─ admin.html          ← Admin dashboard UI
├─ pb-office.html       ← Admin (alternate URL – same as admin)
├─ css/                ← Stylesheets
├─ js/                 ← Frontend JavaScript
│   ├─ main.js         ← Site-wide (cart, contact, social links)
│   ├─ cart.js         ← Shopping basket
│   ├─ checkout-page.js ← Checkout form + PayHere
│   ├─ i18n.js         ← Language switcher
│   └─ translations/   ← en.json, si.json, ta.json
├─ images/             ← Product & hero images
├─ data/               ← outlets.json (branch list)
├─ server/             ← ★ BACKEND ★
│   ├─ server.js       ← Main server (start here)
│   ├─ auth.js         ← Login/JWT
│   ├─ store.js        ← SQLite database layer
│   ├─ seed.js         ← First-run data seeding
│   ├─ payhere.js      ← Payment gateway
│   ├─ admin-api.js    ← Admin routes
│   └─ data/
│       └─ peoples-bakers.db ← Database file (auto-created)
├─ scripts/
│   └─ verify-backend.js ← Health check script
├─ .env                ← Secret settings (YOU create – not in git)
├─ .env.example        ← Template for .env
└─ package.json        ← npm commands
```

5. Server run කරන හැටි — Step by Step

Requirements (අවශ්‍ය දේවල්)

- **Node.js 22.5 or newer** — nodejs.org download
- Project folder extract කරලා තියෙන්න ඕන
- Windows: PowerShell හෝ Command Prompt

Step 1 — Project folder එකට යන්න

```
cd "C:\Users\...\Peoples-Bakers-master"
```

Step 2 — Dependencies install (අලුතු වතාවට විතර)

```
npm install
```

මේක node_modules folder එක create කර dependencies download කරනවා. Internet connection ඕන.

Step 3 — Server start කරන්න

```
npm start
```

Terminal එකේ මේ විදිහ message එකක් පෙන්වන්න ඕන:

```
Peoples Bakers server running at http://localhost:3000
```

Step 4 — Browser එක open කරන්න

Address bar එකේ type කරන්න:

```
http://localhost:3000
```

✓ **Success check:** Home page load වෙනවා, images පෙන්වනවා, cart button work වෙනවා. Terminal window එක **close කරන්න** එපා — server stop වෙනවා.

Other useful commands

Command	What it does
npm start	Server run කරනවා (production mode)
npm run dev	Auto-restart when files change (development)
npm run verify	Backend health check (server running වෙන්න ඕන)
npm run seed	Database re-seed (existing data duplicate නොවෙන විදිහට)

Server stop කරන්න

Terminal එකේ **Ctrl + C** press කරන්න.

6. Important URLs & Pages

URL	Page	Description
http://localhost:3000/	index.html	Home — hero, products, outlets
/order.html	Order Online	WhatsApp, checkout, PayHere
/products.html	Products	Full catalogue + cakes
/account.html	My Account	Profile, order history (login required)
/login.html	Login / Register	Customer accounts
/find-us.html	Find Us	Outlet map & search
/feedback.html	Feedback	Customer reviews
/special-cake.html	Customize Cake	Photo cake requests
/pb-office.html	Admin Panel	Staff dashboard (admin login)
/admin.html	Admin (alt URL)	Same admin panel
/api/health	API check	Server alive ද? JSON response

Default admin login (development)

First run වෙද්දී terminal එකේ admin credentials print වෙනවා. Development default:

Email: admin@peoplesbakers.lk

Password: admin123 (dev only — production වල strong password set කරන්න!)

7. Backend (API) — server එක handle කරන දේවල්

Area	Endpoints	Notes
Auth	POST /api/auth/register, login, logout GET /api/auth/me	JWT cookie — secure login
Orders	POST /api/orders, GET /api/orders POST /api/orders/:id/cancel	Checkout saves here
Products	GET /api/products Admin: POST/PUT/DELETE	Catalog from database
Feedback	POST /api/feedback	feedback.html → here
Payments	POST /api/payments/payhere/checkout POST /api/payments/payhere/notify	PayHere gateway
Outlets	GET /api/outlets, /nearby, /geojson	Branch finder API

Config	GET /api/config	WhatsApp, phone, social URLs
Admin	GET /api/admin/stats, orders, users...	Admin role required

8. Database (SQLite)

Data save වෙනුවෙන් single file එකක්:

```
server/data/peoples-bakers.db
```

- Users, orders, products, feedback, settings — සියල්ල මේ file එකේ
- First start වෙද්දී auto-create + seed වෙනවා
- Reset කරන්න ඕන නම්: server stop → server/data/ folder delete → npm start
- Production: DB_PATH env var use කර persistent disk path එකක් set කරන්න

9. Admin Panel

<http://localhost:3000/pb-office.html> — admin login කරලා:

- New orders view & status update (pending → confirmed → preparing → delivered)
- Products add/edit/delete
- Customer feedback read
- Users list
- Settings (WhatsApp number, business name, ...)
- Kitchen display mode

Admin routes protected — normal customer login වලින් admin data access වෙනම නැහැ.

10. Languages — Sinhala / English / Tamil

Site එකේ top-right corner එකේ language switcher: **EN** | සිං | த

- Translation files: `js/translations/en.json`, `si.json`, `ta.json`
- HTML elements use `data-i18n="key.name"` attributes
- Choice saved in browser `localStorage`

11. Fix කළ Bugs — Summary

Project audit එකකින් හමු වුණ issues fix කරලා තියෙනවා. Main categories:

Critical fixes **FIXED**

Bug	Fix
Checkout stale cart	Submit වෙද්දී cart re-read + event listener
Client price manipulation	Server-side catalog prices only; invalid products rejected
Default admin password	Production: ADMIN_SEED_PASSWORD required; dev-only admin123
Weak JWT secret	Production: JWT_SECRET 32+ chars required; secure cookies

Medium fixes **FIXED**

Bug	Fix
Admin URL 404	<code>/pb-office.html</code> + <code>/admin.html</code> both work
Stale JWT user role	Auth middleware DB refresh
PayHere sandbox default	Production defaults to live unless PAYHERE_SANDBOX=true
PayHere fail cancels order	Failed payment no longer auto-cancels order

Phone discount fraud	Discount only for logged-in users with matching phone
Wrong WhatsApp numbers	All pages use config API + correct default 947228955477
Service worker stale cache	Network-first for main JS/CSS
Sinhala broken text	Translation JSON fixes

Low priority fixes **FIXED**

- Instagram / LinkedIn dead links → real social URLs + config API
- Home page i18n — nav, footer, products, outlets
- Account page i18n — nav + dynamic order text
- Order page FAQ & zone Sinhala translations
- Feedback spam rate limiting
- SQLite write transactions (crash safety)
- Outlets catalog missing file — graceful fallback
- **Mobile hamburger menu** — phone nav compact drawer
- **Slow network image loading** — hero defer + lazy images

12. Environment Variables (.env)

Project root එකේ .env file එක create කරන්න (.env.example copy කර edit):

```
# Development minimum
JWT_SECRET=your-long-random-string-at-least-32-chars
SITE_URL=http://localhost:3000

# Contact (optional – defaults work)
WHATSAPP_NUMBER=947228955477
SITE_PHONE_TEL=+94228955477

# Production MUST set:
# NODE_ENV=production
# ADMIN_SEED_PASSWORD=strong-password
# SITE_URL=https://www.yourdomain.lk
# PAYHERE_MERCHANT_ID=...
# PAYHERE_MERCHANT_SECRET=...
```

Never commit .env to git — passwords/secrets එකේ තියෙනවා.

13. Testing — verify කරන්න

Server run වෙන state එකේ (terminal එක open):

```
npm run verify
```

All checks passed message එක පෙන්වන්න ඕන:

- /api/health, /api/config, /api/outlets
- Admin login test
- Static assets (index.html, images)

Manual test checklist

1. Home page — images, language switch (සිං)
2. Products → cart → order checkout
3. Login → account page → order history
4. Admin login → orders view
5. WhatsApp buttons — correct number
6. **Mobile:** ☰ menu open/close, order flow on phone
7. Hard refresh: Ctrl+Shift+R (cache clear)

14. Common Mistakes — වරදවල්

Mistake	Result	Solution
HTML file double-click	API calls fail, login broken	npm start → localhost:3000
VS Code Live Server use	Backend not running	npm start use කරන්න
Terminal close කරනවා	Site down	Terminal open තියන්න
Port 3000 busy	Server start fail	Old terminal close / PORT=3001 npm start
.env missing JWT_SECRET (production)	Server exit	.env file create කර set කරන්න
Old browser cache	Old JS/CSS load	Ctrl+Shift+R hard refresh

15. Production / Hosting (short)

Live internet වල deploy කරන්න:

- **Node.js supported hosting** ඕන (cPanel Node, Render, Railway, VPS)
- Plain PHP-only / GitHub Pages — **NOT suitable**

- SSL certificate (HTTPS) — PayHere webhooks සඳහා mandatory
- Persistent disk for SQLite database
- Full hosting guide: docs/Peoples-Bakers-Hosting-Guide-Sinhala.pdf

Recommended hosting check: Node.js 22+, 24/7 process, SQLite persistent path, SSL, PayHere notify URL reachable, Singapore DC = OK for SL customers (low latency impact if setup correct).

16. Mobile Responsive — Phone User Experience

Why important? Sri Lanka වල customers බොහෝ දෙනෙක් **mobile phone** එකෙන් site එක use කරනවා. Website එක phone screen එකට fit වෙන්න, fast load වෙන්න, easy navigate වෙන්න ඕන.

16.1 Responsive design — මොකක්ද?

Responsive කියන්නේ screen size එක (phone, tablet, laptop) අනුව layout automatically adjust වෙනවා. Desktop වල wide grid එක, phone වල single column — images, text, buttons overflow නොවී පෙන්වනවා.

Project එකේ:

- `<meta name="viewport" ...>` — phone zoom/layout correct
- `css/style.css` — 30+ media queries (768px, 900px, 1024px breakpoints)
- Grids (products, outlets, footer) → phone වල **1 column**
- Cart side drawer, WhatsApp floating button — touch-friendly
- PWA manifest — home screen add කරන්න පුළුවන්

16.2 Hamburger menu (☰) — NEW improvement

Problem (before): Phone වල nav links wrap වෙලා navbar එක tall, cluttered වුණා.

Solution (now): Screen width **900px** යට:

- ☰ **Hamburger button** — tap කර menu open
- Links side drawer එකක stack වෙනවා (Our History, Products, Order, ...)
- Backdrop tap / link click / Escape key — menu close
- Order Online button compact (icon only on small screens)

Files: `js/main.js` (setupMobileNav) + `css/style.css` (nav drawer styles)

16.3 Slow network / 3G — image speed improvements

Mobile data slow වෙද්දී heavy images site slow කරනවා. Fixes:

Feature	What it does
Hero slide defer	Home page first hero image only loads fast; other 4 slides load later (staggered)
Hero preload	First hero image preloaded for faster first paint
Lazy images	Below-fold <code></code> tags get <code>loading="lazy"</code>
Product card backgrounds	Home page product photos load when user scrolls near them

Files: js/main.js (setupProgressiveImages)

16.4 Mobile UX checklist — test කරන්න

- 1. Chrome DevTools → **Ctrl+Shift+M** (mobile view) හෝ real phone
- 2. ☰ menu open → link tap → menu close
- 3. Home scroll — products images appear when scrolling
- 4. Cart → checkout form fill (phone keyboard)
- 5. WhatsApp FAB button tap
- 6. Language switch ඕං on phone
- 7. Hard refresh **Ctrl+Shift+R** after updates

Live hosting වලින්: Mobile responsive CSS/JS server එකෙන් serve වෙනවා — domain live කරපු පස්සේ phone වල same experience. Hosting guide: Peoples-Bakers-Hosting-Guide-Sinhala.pdf (Desktop එකේ).

Quick Reference Card

Task	Command / URL
Install	npm install
Run server	npm start
Open site	http://localhost:3000
Admin	http://localhost:3000/pb-office.html
Verify	npm run verify
Stop server	Ctrl+C in terminal
WhatsApp default	+94 22 895 5477 (947228955477)
Database file	server/data/peoples-bakers.db
Mobile menu	☰ button (screen < 900px)
Hosting PDF	Peoples-Bakers-Hosting-Guide-Sinhala.pdf